

Toward the "Sound Barrier"

Huge resources were poured into aircraft development during World War II because having faster machines than the other side could make a significant difference. Rocket-powered interceptors flew at previously unknown speeds, and piston-engined planes approached the speed of sound when diving. The research was not wasted after the war because swept wings, jet engines, ramjets, and rockets took their place and the speed of sound (Mach 1) was easily exceeded.

▷ Supermarine Spitfire PR MkX 1944

Origin UK

Engine 2,375 hp Rolls-Royce Griffon 69 liquid-cooled V12

Top speed 483 mph (778 km/h)

This aircraft was based on the Spitfire but with new laminar-flow wings and a new fuselage to combat instability. The Spiteful was overtaken by jet fighters; 19 were built (2 prototypes and 17 production).



▷ Supermarine Spitfire PR MkX 1944

Origin UK

Engine 1,655 hp Rolls-Royce Merlin 77 supercharged liquid-cooled V12

Top speed 417 mph (671 km/h)

The thin-winged Spitfire had the highest limiting mach number of any WWII piston-engined aircraft. A fully instrumented MkXI, similar to the aircraft shown, reached speeds of 606 mph (975 km/h), or Mach 0.891—nearly nine-tenths the speed of sound—in a dive.



▽ Messerschmitt Me 163 Komet 1944

Origin Germany

Engine 3,750 lb (1,701 kg) Walter HWK 109-509A-2 liquid-fuel rocket

Top speed 596 mph (960 km/h)

This was the only rocket-powered aircraft to see active service, relying on its ability to overtake high-altitude bombers and make one or two diving passes before its engine cut out. In a dive it reached 698 mph (1,123 km/h).



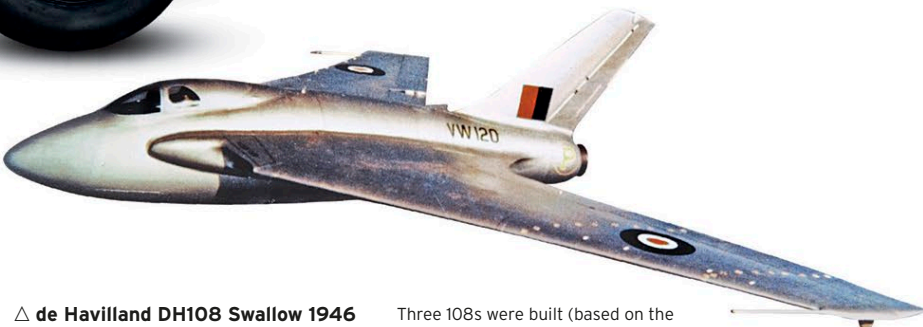
△ Gloster Meteor F4 1944

Origin UK

Engine 2 x 3,500 lb (1,588 kg) thrust Rolls-Royce Derwent V turbojet

Top speed 616 mph (991 km/h)

The world's first production jet set new world speed records after WWII, raising it from 469 mph (755 km/h) to 606 mph (975 km/h) in 1945 and on to 616 mph (991 km/h) in 1946. It also set rate of climb and endurance records.



△ de Havilland DH108 Swallow 1946

Origin UK

Engine 3,738 lb (1,696 kg) thrust de Havilland Goblin 4 centrifugal compressor jet

Top speed 605 mph (974 km/h)

Three 108s were built (based on the Vampire jet fighter) to test tailless swept-wing handling; all three crashed fatally, but not before this final example had set a new world speed record for a 62-mile (100-km) circuit.